

Fabio Battaglia

LEAD SOFTWARE ENGINEER ·

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Summary

Software Engineer with proven expertise in designing cutting-edge systems across Aerospace, Biometric Security, and Travel sectors.

Led a **2-year European Space Agency (ESA)-co-funded** project from design to deployment.

Master's Degree in Automation Engineering with a focus on control systems, image processing and robotics. Skilled in backend development, hardware integration (microcontrollers and embedded systems), data handling across SQL/NoSQL databases, with experience in both on-premise and cloud deployments (AWS). Combines deep technical expertise with cross-functional collaboration to drive innovation in the aerospace domain.

Work Experience

Spire Global

Luxembourg, Luxembourg

SENIOR SOFTWARE ENGINEER

Dec. 2022 - Present

- Develop, deploy, and maintain core services for the Space Services unit, enabling effective management of custom satellite constellations.
- Led and delivered a 2-year development project co-funded by the European Space Agency (ESA), driving design, implementation, and integration phases.
- Architected microservices using **FastAPI, Flask, PostgreSQL, Terraform, Docker, Kubernetes, GitHub CI/CD actions**, minimizing manual deployments and accelerating release cycles.
- Collaborated with cross-functional teams—mission operators, hardware engineers, and product managers—to define requirements, establish timelines, and ensure successful project outcomes.
- Established robust monitoring and alerting solutions (e.g., Prometheus, Grafana), improving platform reliability and reducing mean time to recovery (MTTR).

Remark AI

London, United Kingdom

LEAD SOFTWARE ENGINEER

Sep. 2020 - Oct. 2022

- Architected and developed an application for thermal scanning, mask detection, and **face recognition** using thermal or standard network cameras, improving on-site health and security measures.
- Created microservices leveraging **Flask, FastAPI, C++, MongoDB, MySQL, GraphQL, Kafka, Traefik, Docker, Kubernetes** and deployed them via GitLab CI/CD (on AWS and on-premise).
- Implemented and optimized AI pipelines (e.g., **DeepStream** in C++/Python) for real-time face recognition, collaborating with data scientists to refine model accuracy and speed.
- Onboarded and mentored backend and frontend engineers, including code reviews, pair programming, and knowledge-sharing sessions.
- Oversaw migration of legacy Python 2 backends to Python 3 and async implementations, reducing technical debt and boosting system throughput.
- Developed an RTSP re-streaming server using GStreamer to broadcast local files over RTSP in real time, enhancing the scalability of video analytics pipelines.
- Coordinated sprint planning and Agile ceremonies (Scrum/Kanban), fostering team alignment and consistently delivering on schedule.

Amadeus

London, United Kingdom

SENIOR SOFTWARE ENGINEER

Sep. 2017 - Sep. 2020

Description:

- Global provider of IT services for travel, hotels, and airlines.
- Developed features for Amadeus Altéa Departure Control System (DCS), managing passenger flow from check-in to boarding.
- Focused on regulatory requirements, ensuring seamless data provision for authorities and passengers.

Activities:

- Designed and developed server applications using **C++11**.
- Coordinated an urgent delivery for interactive data submission to Chinese authorities, from development to production in 3 months.
- Implemented a configuration system to streamline deployment across multiple environments.
- Drove a proof of concept using **Drools** with Java, OpenShift, and MongoDB for rules-based decision-making.
- Mentored junior developers and interns, providing **supervision** and **project planning** support.

XiD Technologies

Singapore

SOFTWARE ENGINEER

Feb. 2015 - Aug. 2017

- Developed solutions integrating 3rd-party face/fingerprint devices into **XiD FaaS** (e.g., Hanvon, ZK, Suprema).
- Implemented an integrated fingerprint/RFID security system and CCTV for **Changi Airport Terminal 5**, enabling real-time alerts, anti-passback, and work permit expiration features.
- Created computer vision algorithms (HAAR, ASM, histogram-based scoring) in MATLAB/C++ for biometric analysis.
- Led embedded software development for XiD Revo Face 1.0 and 1.5, using ARM 9-based architecture with custom hardware integrations.
- Transitioned to off-the-shelf RTSP cameras and designed an orchestrator in C# to manage data streams for face recognition.
- Built an Android-based attendance and verification app using fingerprint/RFID (Android Studio, Java, MySQL).

ENEACONTROL SYSTEMS ENGINEER INTERNSHIP

Rome, Italy

Jun. 2014 - Oct. 2014

- Investigated and modeled thermal behavior of a Tokamak liquid lithium limiter for nuclear fusion research.
- Developed a real-time monitoring tool in **MATLAB** and **C++** to analyze infrared thermal camera data and define a mathematical heat-distribution model.

High School C. Marchesi

Catania, Italy

Mar. 2014 - Jun. 2014

ROBOTICS TEACHER

- Taught an EU-PON-funded course in robotics to a class of 20 high school students, covering theory and hands-on projects.
- Introduced fundamental concepts in electronics, programming, and control systems, with practical lab exercises.
- Guided students in building an auto-orienting PTZ camera (Arduino, servomotors, sensors) and a 2-wheel mobile robot for obstacle avoidance.

Education

University of Catania

Catania, Italy

Apr. 2012 - Oct. 2014

MASTER'S DEGREE IN AUTOMATION ENGINEERING & CONTROL OF COMPLEX SYSTEMS

- Focus on **Control Systems, Bioengineering, Image Processing, Industrial Automation, Sensors & Robotics**.
- Developed projects involving humanoid robots (image processing), autonomous mobile robots (infrared sensors), PLC-based simulations (parking lot), inertial sensor integration, and CNN development on STM32 boards.
- Thesis: **"Frascati Tokamak Upgrade: New Trends in Model Identification of Specific Thermal Phenomena"** (Supervisors: Prof. Luigi Fortuna, Dott. G. Mazzitelli, Ing. A. Buscarino).
- Graduated with **110/110 cum laude** and special mention.

University of Catania

Catania, Italy

Oct. 2008 - Mar. 2012

BACHELOR'S DEGREE IN COMPUTER ENGINEERING

- Studied **Mathematics, Physics, Electronics, Networking, Software Engineering, Operating Systems, Programming (C/C++/Java/Assembly), Databases, Control Systems**.
- Thesis: **"NetFPGA Architecture and Hardware Description"** (Advisors: Prof. G. Schembra, Dott. D. Reforgiato).

Languages

Italian (native), English (C2), French (A2)

Skills

Programming Languages	C++11+, Python 3, TypeScript, Rust, C, C#, Java, Matlab, $\LaTeX$
Python Ecosystem	FastAPI, Flask, Pydantic, SQLAlchemy, PyTest, MyPy
DevOps & Cloud	Docker, Kubernetes, Terraform, Jenkins, GitLab/GitHub CI/CD, AWS (S3, EKS, Lambda, RDS)
Databases	PostgreSQL, MySQL, MSSQL, Elasticsearch, MongoDB
Frameworks & Tools	OpenCV, TensorFlow, GStreamer, FFmpeg, Android SDK, Boost, GraphQL, Kafka
Protocols & Formats	HTTP/1.1–2, gRPC, RTSP, JSON, YAML, XML, CSV, Protocol Buffers
Version Control & PM	Git (GitLab, GitHub), Mercurial, SVN; Agile (Scrum/Kanban), Waterfall
Electronics & Embedded	STM32, Arduino, Raspberry Pi, PCB design, Siemens PLCs, servo/DC motors